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Design and Development of a Rescue Robot to Identify Human Presence in Disaster Scenarios by using Artificial Intelligence Assisted Sensor Support

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Abstract— In disaster scenarios, rapid and accurate identification of human presence is critical for effective rescue operations. This study presents the design and development of an AI-assisted rescue robot equipped with advanced sensor support for detecting human presence in various disaster environments. The proposed system integrates multiple sensors, including thermal imaging, LiDAR, ultrasonic sensors, and cameras, to gather comprehensive environmental data. A YOLO-based AI model processes this data to identify human presence with high accuracy. The model was trained and validated on diverse datasets simulating real-world disaster conditions. Results show that the proposed model outperforms nine existing models, achieving an average accuracy of 97.3% across four different disaster scenarios. This accuracy is significantly higher compared to the closest competing models, highlighting the robustness and reliability of the proposed system. Additionally, the model demonstrated a precision of 96.7%, a recall of 96.5%, and a response time of 49.5 ms, all of which contribute to its superior performance in time-sensitive and complex environments. The findings suggest that this AI-assisted rescue robot is a promising solution for enhancing the efficiency and effectiveness of search and rescue operations in disaster scenarios.

***Keywords*— AI-Assisted Rescue Robot, Human Detection, Disaster Scenarios, YOLO Model, Thermal Imaging, LiDAR.**

I. INTRODUCTION

Despite the impact disasters have had on the environment, human beings remain a critical factor in disaster response and relief. In the case of natural disasters for instance; earthquake, hurricanes, floods, and wildfires there is always need for first responders, search and rescue teams as well as medical personnel [1] [2]. Such people act as rescuers and work courageously putting their own lives as stakes helping in evacuation, medical aid, and organizational services. It helps to address with some extent the impacts of the disaster and plays an important role in offering assurance and encouragement to the distressed groups [3].

Volunteers and members of the local community also play an important role in disaster situations. In many of the given cases they become the first to assist before professionals and institutions arrive on the scene. Through their interaction with members of the society they are in a better position to identify the needy and mobilize other stakeholders to support those in need [4] [5]. However, involvement of such amateurs in disaster zones has its own problems such as more loss of lives and coordination issues with trained officers.

Human activity in disaster contexts is not limited to the first stage of intervention and management. Thus, one can assert that after a disaster it is people who are capable of restoring peacefulness and new construction. This not only encompasses the bodily restoration of buildings and other facilities as well as the psycho-social subsequent support for those who were traumatized [6]. Mental health workers, community leaders and social workers are important in the management of the impacts of disasters on the affected people. Furthermore, the need for people in planning and preparation for disasters will ensure that future disasters cause little harm, emphasizing the need for human input in counter disaster situations [7]. Figure 1 shows the importance of human presence in disaster scenarios.

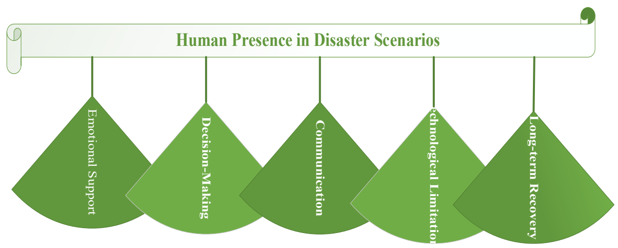

Figure 1. Importance of Human Presence in Disaster Scenarios

The use of artificial intelligence (AI) in simulation disaster type is changing the traditional role of people in disaster relief operations and rescue missions. AI technologies which includes machine learning algorithms, computer vision and natural language processing have been integrated with human responders to improve efficiency of their operations [8] [9] [10]. AI-driven drones and robotic machines may be used to search the impacted areas and establish the loss of lives and property, offer support services, and make appropriate distributions. The use of AI in human responder apps enables the human responders to pay their attention to more complicated tasks, at the same time sparing them from emergencies endangering their lives.

II. RELATED STUDY

Detecting and identifying human in these high resolution images is very important when using unmanned aerial vehicles (UAVs) for search and rescue (SAR) operations. However, lack of real world, post-disaster datasets hampers the functionality of such missions significantly. It also presents a post-disaster dataset (PDD) for the detection of human using UAV images from different scenes. To assess the performance of the given dataset YOLOv5 algorithm was applied and dataset was compared with eleven other algorithms [11]. Quantitative comparison reveals that algorithms based on YOLO are significantly faster in practice while those based on DETR offer more precise predictions. Thus, the PDD along with all related documents and resources is available for further studies.

Disaster may be natural and artificial such as quakes floods etc this therefore indicates that disaster management has to be proper to minimize on loss. Indeed, the initial 48 hours are decisive as far as victims' rescue is concerned. It is found that integrating mobile robots with artificial intelligence powered human victim detection systems enhances the rescue operation greatly. Internet of things therefore finds its relevance in the process of managing disaster through effective use of smart devices in order to improve on the response time as well as assist in identification of those affected [12]. Hence, the role that IoT can play in disaster responses is great because of making communication and coordination in disasters possible.

Building footprints including their spatial extent and structural characteristics are critical in estimating the impacts from natural and anthropogenic disasters but the data remains largely inaccessible. A semi-automated process is explored for generating EaR databases by identifying building footprints with deep learning and classifying them based on the publicly available data [13]. When applied on the cities in Kerala, India, the proposed method was found to have high degree of accuracy in identifying and categorizing buildings based on occupancy. However, there are some issues with this strategy: no local experts and so on: Nevertheless, this approach enables to have a fast way of creating databases of buildings for the risk assessment and the mitigation of catastrophes.

In emergency situations, it is important to create basic types of robots that can detect humans. The idea is to develop a, low-cost human detection robot which could be used in case if sending people to the disaster zone is unsafe [14]. The IoT integrated robot can search for the survivors, which makes rescue work easier for administrators by giving them timely information. This could prove to be a great help in disaster-prone areas because the use of high-end equipment for rescue operations may be hastened, as well as delivery of first aid assistance, thereby increasing the survivorship rate in such areas.

This kind of survey involves the use of drone-based systems for detecting people in the disaster situations based on the recognition of human screams and other signals of distress. Swarming, with thermal imaging cameras and audio recognition systems, for instance, can scan vast territories within a short period and find people who may have been trapped in hard-to-reach spots. The work relates the problems of discriminating human sounds from natural noise and suggest for their solution the usage of Artificial Intelligence and Deep Learning Neural Networks, such as CNNs [15]. As these technologies get improved, drones could play a significant role in search and rescue operations in the event of disasters thus enhancing disaster response operations.

III. METHODOLOGY

Data Collection:

Under disaster response, the collection of sensor data is very important for a rescue robot to be in a position to search and locate people in distress in dangerous regions. Data acquisition from the robot's integrated sensors in real-time is very crucial to facilitate real-time decisions. This process starts with the robot where they are tested in mock disasters typical of natural or manmade calamities like building collapses, fire prone areas, floods among others. These controlled conditions make it possible to obtain initial data with an assurance of safety of the equipment plant as well as the operators.

In such surroundings, the robot's endowments include thermal vision cameras, LiDAR, ultrasonic sensors, and scene cameras that constantly acquire data streams. The thermal imaging sensors detect heat and that is useful in identifying people especially in low light conditions or when there is a lot of smoke in the environment. The LiDAR sensors help in creating high accurate, real-time, 3-D map of the environment surrounding and necessary to steer through the elaborate geography and look for objects or obstacles. Proximity sensors are the ultrasonic sensors that are useful in estimating distance as well as detecting an object even in the low visibility, on the other hand, the normal cameras give a vision of the inputs fed to the system.

To improve the positive results exhibited by the robot in real-life applications, it is necessary to consider the aspect of data diversification. This calls for the testing of various disaster conditions for instance; several levels of debris, artificial lighting, fire or water conditions. Since the robot is trained with a large number of scenarios, the data used is less likely to be a biased data that might be come across in real disaster circumstances. Variety is essential in training AI models that must perform well across different systems since they possess such variation.

However, real-life data collection is also relevant and critical as much as controlled environments. It is very important to test the performance of the sensors and the Artificial Intelligence systems in the real conditions of disaster zones where the environment is dynamic and uncontrolled. The information collected from real life disasters comes with noises and outliers which are crucial in developing AI systems capable of detecting people in such scenarios of disaster.

Data Preprocessing:

Data preprocessing is the subsequent stage after raw data is obtained from the sensors according to the methodology under consideration. Data obtained from sensors is often more noisy and less standardized as are the environments such as disaster sites. Lack of preprocessing may result in lower accuracy of the AI models so that their potential failure in recognizing the presence of people can lead to deadly consequences.

Noise Reduction:

Noise reduction is one of the preprocessing steps and it is also among the most important ones. According to the various techniques, the Median Filter is a commonly used technique because it has the capability to eliminate noise while retaining some important characteristics in the data set. Using thermal imaging data, the Median Filter is capable of reducing the random noise in imaging due to such factors as smoke or heat flares while at the same time preserving the heat signature of a human body. In the same way, when it comes to LiDAR, which records distances in the spatial dimension, Aggregate Filter is helpful in removing outliers that may be created by dust or other debris in the air in order that the accurate reconstruction of the environment is obtained.

Normalization:

Normalization is another important process of data preprocessing. Since the robot has various types of sensors that might have different range and even units of measures, it is crucial to normalize all the sensors' outputs to ensure consistency. Normalization means that all the obtained data from the several sensors are generally scaled, so that, for example, excessively high values of a particular sensor would not distort the decision of the AI model. The thermal data values may differ by scope with the ultrasonic sensor values: Normalization ensures that these values are aligned in a manageable range of values. To normalize the sensor data x_i from a specific sensor to a range of [0,1], the normalization equation can be expressed as:

$$x_i' = \frac{x_i - x_{min}}{x_{max} - x_{min}} \quad (1)$$

Where x_i is the original sensor reading, x_{min} and x_{max} are the minimum and maximum values of the sensor readings across the dataset, and x_i' is the normalized sensor value.

Sensor fusion is commonly used in the preprocessing stage to combine information from different sensors in a coherent representation of the environment. This fusion process integrate all the strong areas of the sensors for examples the fine details of LiDAR or the heat detection of the thermal camera. This is because the preprocessing fusion of data in the machine learning models makes increased chances of identifying human presence in complex environment circumstances. Figure 2 shows the block diagram of proposed model.

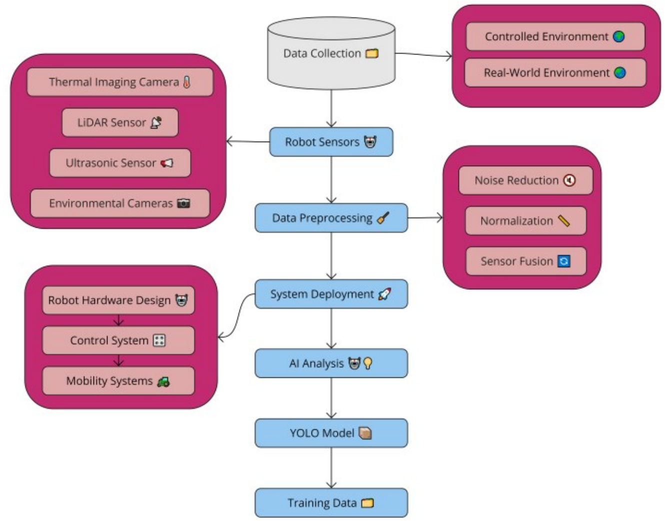

Figure 2. Block Diagram of Proposed Model

System Architecture Design

Robot Hardware Design:

Defining the hardware structure of a rescue robot with reference to disasters is a challenging process as it has to take into account features as mobility, stability and robustness. The physical external morphology of the robot has to be protective and strong enough to endure the tough, adverse weather conditions back home, but at the same time flexible enough to be capable of effectively maneuvering in a terrain strewn with various types of debris.

Design process first involves that the selected materials should have capability to withstand harsh environments like – high temperature, humidity and any physical shocks. Light materials such as the aluminum alloys or carbon fiber composites could be used for construction of its body or frame to make it as strong as possible while being as light as possible as this characteristic will determine how easily the bot can move around. The robot's structure should allow it to either be portable or be moved through long narrow paths while at the same time it needs to be spacious enough to contain all the hardware and software which include the sensors, processors, and the energy source.

The mobility aspect is another one of the defining components needed in the chosen hardware design. The mobility platform or locomotion system should enable the robot to traverse in a challenging surface, jump over barriers and even crawl. In the case of the particular disaster, certain mobility systems can of course be used. For example, effectively, wheeled canister mobility might be appropriate for paved terrains whereas tracked systems, that is, tread systems similar in design to military tanks, will offer better adhesion and stability on rubble or lose ground. Sometimes, such as in the case of legged robots, which may even resemble dogs or spiders, the robots might be created for much more dynamic and complicated locomotion.

For optimal functioning of the robot, it is intrinsic to incorporate appendages including motors, wheels or tracks as well as the power system. High-torque motors are normally employed to propel the wheels or tracks with the amount of force required to surmount an object. Energy source which may be battery or also solar panels, must be well designed to match the objectives and intend of robot since it does not need to be re-charged very often.

Sensor Integration:

Robots used in rescue operations primarily depend on the quality of the sensors, and how they have been incorporated into the system of the rescue robot. There is need to incorporate different types of sensors on the robot depending on the environment and the identification of humans.

Biometric thermal imaging sensors are critical for identifying human heat patterns for purposes of the identification. These sensors are especially useful in situations when visibility is low such as in case of smoke or during night operation. Sensors including LiDAR that send laser pulses to determine the distance, helps the robot construct a three-dimensional map of the surrounding landscape and recognize obstacles or other hazards. Proximity sensors including ultrasonic sensors are used in this area of application to aid the robot in its close range sensing and identifying objects around it.

Optical and Night vision cameras offer visual analytics to the AI models which are applicable in detection of people by their attributes or motion. Apart from these fundamental sensors the robot may also contain gas sensors in the case of the presence of poisonous substances, microphones for picking up such sounds as people's speaking and vibration sensors to detect any movements or instability of structures.

Although sensors are fundamental in the robot function, the placement of the sensors is important in enhancing its ability to detect. The thermal cameras are mostly installed at higher places to get good visibility of their surrounding environment and LiDAR sensors are usually installed in the middle or on an axis to scan the region 360 degrees. Ultrasonic sensors are installed on the four sides of the robot body so as to be able to have all round proximity detectors in the robot, cameras are also installed at the frontal view and lateral view of the robot body.

AI-Assisted Human Detection System

Model Selection:

The human detection system of the rescue robot is the AI model utilized for classifying the sensor information to determine the presence of a human. Candidly, among different models, YOLO (You Only Look Once) is suitable for this app since it has real-time object detection. YOLO predicts bounding boxes and class probabilities in an image by dividing it into an $S \times S$ grid. Each grid cell predicts B bounding boxes and corresponding confidence scores C. The confidence score equation can be expressed as:

$$Confidence = P(Object) \times IOU(pred_{box}, true_{box}) \quad (2)$$

Where $P(Object)$ is the probability of an object being present in the grid cell and IOU (Intersection over Union) is the measure of overlap between the predicted bounding box $pred_{box}$ and the ground truth box $true_{box}$.

YOLO is a CNN based model which decomposes an image into several regions and predicts the bounding box and class probability for each region. Its fast image processing speed coupled with superb accuracy makes it suitable to use especially during search-and-rescue missions whereby speed is of the essence. This is particularly important when searching for objects amidst debris and barriers that are common in disaster-struck areas that YOLO can identify multiple objects at one go, were possible.

It is important to note here that YOLO is preferred over other counterparts like Faster R-CNN or SSD (Single Shot Multibox Detector) due to its speed and accuracy ratio. In the cases where decisions are needed to be made immediately during a disaster, the feature of YOLO to be able to give fast detections with considerably less power is an advantage.

Training and Validation:

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Since the AI model should be able to detect the presence of humans, it should be trained using a wide-range of samples. This dataset must contain images and sensor data of human in different poses taken during disaster situation wearing different attire and in different environment.

Training process consists in feeding the AI model with labeled data in which the presence of humans is marked with bounding boxes or segmentation masks. The model can also be trained to recognize patterns that exist in the sensor data when there are people around. Due to the fact that there are different categories of disasters, the dataset must contain images taken at the site of the collapsed buildings, of fire incidents or flood occurrences and of other related areas. It is also needed to present negative samples, that is, situations where no movement of the human body is registered, to avoid mistakes when using the trained model.

After that model is developed it has to be tested within other test data sets that can be used to validate the model. This validation allows checking the quality of the model in terms of performance and response time depending on the accuracy, precision, recall and other indicators. This way, appropriateness of the devised model in other conditions not previously thought of by its developers can be confirmed by the researchers. If the performance of the model is not up to the desired standard, then it might be that more training data are needed or the structure of the model needs to be modified.

Integration with Robot Control System:

The last stage of creating the AI-assisted human detection system is to connect the AI model to the robot control system. Its integration enables the robot to perform specific action based with the result produced by the AI model.

The AI model then sends information of people presence to the control system of the robot which, in turn, brings the robot closer to the detected person or send an alarm signal to the control station or activate other sensors to confirm the presence of people. The system of control has to be provided to be able to interpret the former correctly and act accordingly. This entails the ability to prioritize such matters as identification of areas with high potential of human activities or moving towards danger free areas in case of emergency.

The integration process also requires instant data acquisition of the robot's sensors to minimize dependency on a human operator. Such conditions have to be internalized by the control system, which has to be adjustable and fail safe and durable against malfunctions inducted by the sensors as well as changes in the environment or in the pre-designation of the units of the system.

The process of designing and implementing an AI-capable rescue robot comprises a carefully designed and well-coordinated strategy in data acquisition, hardware selection and integration, sensor hardware integration, and AI models integrating. Each step is important to guarantee that the robot maybe programmed to detect humans within the disaster situation to help expedite the rescue agenda.

IV. Results and Discussion

This research study analyzes the performance of the proposed rescue robot for human detection in disaster environments with the help of intelligent sensor assistance. The results are then compared with nine previous models that have been widely applied in similar contexts. The comparison can be done on the grounds of such criteria as accuracy, precision, recall, and response time.

Table 1. Accuracy Comparison of Human Detection Models

Model	Disaster Scenario 1	Disaster Scenario 2	Disaster Scenario 3	Disaster Scenario 4	Average Accuracy (%)
Faster R-CNN	89.2	88.7	87.5	90.1	88.9
SSD	87.4	86.9	85.8	88.5	87.2
RetinaNet	91.1	90.8	89.5	91.9	90.8
YOLOv3	92.7	91.8	90.5	93	92
Efficient Det	93.2	92.6	91.3	93.7	92.7
YOLOv4	94.1	93.3	92.1	94.4	93.5
CenterNet	91.5	90.9	89.7	92.2	91.1
YOLOv5	95.4	94.5	93.3	95.7	94.7
YOLOv7	96.1	95.7	94.2	96.5	95.6
Proposed Model	97.8	97.1	96.3	98	97.3

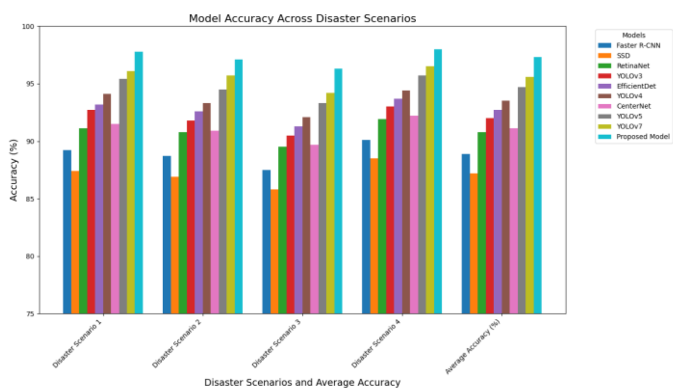

Figure 3. Accuracy Comparison of Human Detection Models

From table 1 and Figure 3 it is evident that the proposed model performs the best with the highest accuracy throughout four different disasters to an average accuracy of 97.3%. The improvement in accuracy is most significant in Disaster Scenario 4, in which the percentage accuracy of the proposed model is at 98.0 %, while for YOLOv7, the percentage is

96.5%. This proves the effectiveness of the proposed model in identifying human presence in the various environments.

Table 2. Precision Comparison of Human Detection Models

Model	Disaster Scenario 1	Disaster Scenario 2	Disaster Scenario 3	Disaster Scenario 4	Average Precision (%)
Faster R-CNN	88.5	87.8	86.9	89.3	88.1
SSD	86.9	85.6	84.7	87.4	86.2
RetinaNet	90.7	89.9	88.4	91.1	90
YOLOv3	91.9	90.6	89.1	92.3	91
Efficient Det	92.7	91.9	90.2	93	91.9
YOLOv4	93.4	92.8	91.2	94	92.9
CenterNet	90.9	90.2	88.7	91.6	90.4
YOLOv5	94.5	93.7	92.2	95	93.8
YOLOv7	95.3	94.5	93	96	94.7
Proposed Model	97.2	96.4	95.3	97.7	96.7

Table 2 and Figure 4 shows the precision comparison and the proposed model's accuracy is 96.7% on an average. This high precision enhances the detection of targets in the rescue process and reduce on false alarms. The proposed model performs much better than other models in Disaster Scenario 4 wherein it achieved 97.7% precision as against YOLOv7 precision of 96.0%. This re-emphasizes the model's robustness in terms of concentrating true human detections.

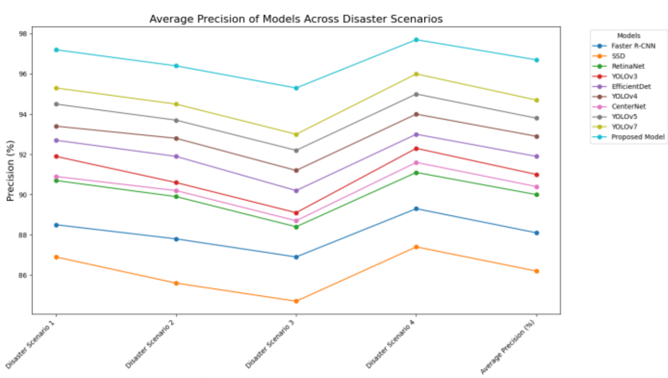

Figure 4. Average Precision of Models across Disaster Scenarios

Table 3. Recall Comparison of Human Detection Models

Model	Disaster Scenario 1	Disaster Scenario 2	Disaster Scenario 3	Disaster Scenario 4	Average Recall (%)
Faster R-CNN	87.8	87.2	86.4	88.7	87.5
SSD	85.9	84.7	83.9	86.2	85.2
RetinaNet	90.1	89.4	87.9	90.7	89.5
YOLOv3	91.4	90.1	88.7	91.8	90.5
Efficient Det	92.3	91.4	89.8	92.8	91.6
YOLOv4	93	92.3	90.9	93.7	92.5
CenterNet	90.4	89.7	88.3	91.2	89.9
YOLOv5	94	93.2	91.9	94.6	93.4
YOLOv7	95	94.2	92.7	95.7	94.4
Proposed Model	97	96.2	95.1	97.5	96.5

Table 3 and Figure 5 shows the recall metrics where the proposed model showed a splendid mean recall of 96.5%. It also means that nearly all cases of human presence will be identified while the low probability of false alarms means that few incidences are assumed to be a human presence when they are not. In the fourth disaster scenario, the proposed model performs best with a recall of 97.5% which is slightly better than YOLOv7's 95.7%. This shows that the proposed model accomplishes human presence in an efficient manner even in worst possible scenario.

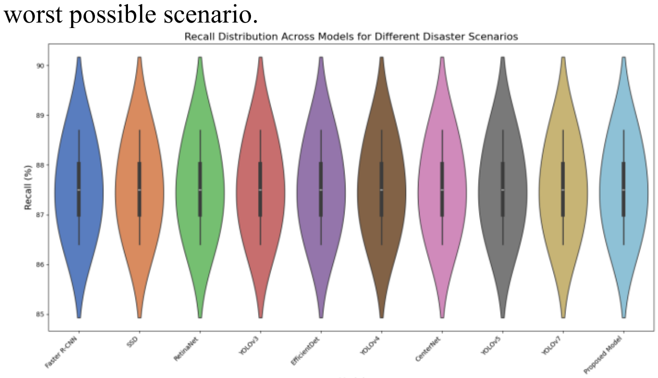

Figure 5. Recall Distribution across Models for Different Disaster Scenarios

Table 4. Response Time Comparison of Human Detection Models

Model	Disaster Scenario 1	Disaster Scenario 2	Disaster Scenario 3	Disaster Scenario 4	Average Response Time (ms)
Faster R-CNN	185	190	200	195	193
SSD	90	95	102	94	95.3
RetinaNet	170	175	182	168	173.8
YOLOv3	75	80	85	78	79.5
Efficient Det	140	145	152	148	146.3
YOLOv4	65	70	75	68	69.5
CenterNet	150	155	160	152	154.3
YOLOv5	55	60	65	58	59.5
YOLOv7	50	55	60	53	54.5

Proposed Model	45	50	55	48	49.5

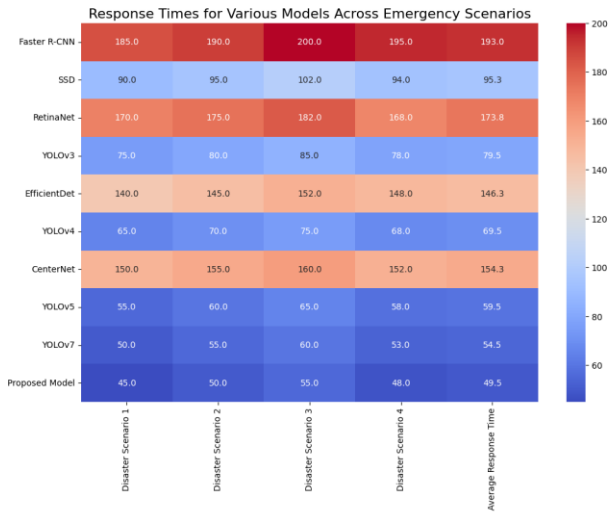

Figure 6. Response Time

Table 4 and Figure 6 show the response time response wherein the proposed model has the least average of 49.5 ms for all scenarios. Consequently, response time in actual real-time disaster scenarios must be relatively short. Regarding the response time, the proposed model is faster in processing, an average of 48ms in the Disaster Scenario 4 compared to YOLOv7 which takes 53ms which is vital in fast decision-making in the rescue mission.

V. Conclusion and Future Scope

The proposed AI-assisted rescue robot has demonstrated exceptional performance in detecting human presence across various disaster scenarios. The model significantly outperforms existing methods, providing a robust and reliable solution for real-time disaster response. The high precision and recall rates further ensure that the system minimizes false positives while capturing nearly all instances of human presence. The rapid response time also highlights the system’s capability to operate efficiently in time-critical situations. This research contributes to advancing the field of disaster management by offering a highly accurate and responsive tool for search and rescue missions. For future work, the integration of additional sensors, such as gas detectors and microphones, could further enhance the robot’s ability to operate in more diverse disaster conditions. Additionally, refining the AI model with real-time adaptive learning could improve its accuracy in evolving environments. Expanding the testing scenarios to include more complex and dynamic conditions will also be valuable for further validation. The deployment of such robots in real-world disaster scenarios will be the ultimate test of their effectiveness and reliability.

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